

~~Examen 15/08/2022~~

1

$$T(n) = \begin{cases} C_1 & n = 1 \\ 2T(n/2) + C_2 + \sum_{j=1}^n C_3 & n > 1 \end{cases}$$

Paso 1 $2T\left(\frac{n}{2}\right) + C_2 + n \cdot C_3$

Paso 2 $2\left[2T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right) \cdot C_3 + C_2\right] + n \cdot C_3 + C_2 =$

$$= 2 \cdot 2T\left(\frac{n}{2}\right) + 2 \cdot \left(\frac{n}{2}\right) \cdot C_3 + 2 \cdot C_2 + n \cdot C_3 + C_2 =$$

$$= 2^2 T\left(\frac{n}{2^2}\right) + n \cdot C_3 + 2 \cdot C_2 + n \cdot C_3 + C_2 =$$

$$= 2^2 T\left(\frac{n}{2^2}\right) + 2 \cdot C_3 \cdot n + 3 \cdot C_2$$

Paso i $2^i T\left(\frac{n}{2^i}\right) + i \cdot C_3 \cdot n + (2^i - 1) \cdot C_2$

Caso base $\frac{n}{2^i} = 1 \rightarrow n = 2^i \rightarrow \log_2(n) = i$

Reemplazo el valor de i en el paso general

$$2^{\log_2(n)} T\left(\frac{n}{2^{\log_2(n)}}\right) + \log_2(n) \cdot n \cdot C_3 + (2^{\log_2(n)} - 1) \cdot C_2 =$$

$$= n \cdot T(1) + \log_2(n) \cdot n \cdot C_3 + (n - 1) \cdot C_2 = n \cdot C_1 + \log_2(n) \cdot n \cdot C_3 + n \cdot C_2 - C_2$$

$$= n \cdot C_1 + \underbrace{\log_2(n) \cdot n \cdot C_3}_{\text{Término dominante}} + n \cdot C_2 - C_2 = O(n \cdot \log_2(n))$$

Término dominante

2) Calcular el $O(n)$ por definición

$$T(n) = n^3 + 2^n + 1000$$

3)

1 hora $\rightarrow$ 1000

$$C \cdot \log_{10}(n) = \text{tiempo}$$

3 horas $\rightarrow$ 1000.000.000

$$C \cdot \log_{10}(n) = 1$$

$$C \cdot \log_{10}(n) = \text{tiempo}$$

$$C \cdot \log_{10}(1000) = 1$$

$$\frac{1}{3} \cdot \log_{10}(n) = 3$$

$$C \cdot 3 = 1$$

$$\log_{10}(n) = 3 \cdot 3$$

$$C = \frac{1}{3}$$

$$\log_{10}(n) = 9$$

$$n = 10^9$$

$$n = 1.000.000.000$$

4) 1) opción C $\rightarrow T(n) = 1 + 2T(n/2)$

4) 2) opción a $\rightarrow O(2^n \cdot \log(n))$

$$(2^n + 2) \cdot (\log(n) + 5) = 2^n \cdot \log(n) + 2^n \cdot 5 + 2 \cdot \log(n) + 2 \cdot 5 =$$

$$= 2^n \cdot \log(n) + 2^n \cdot 5 + 2 \cdot \log(n) + 10$$

4) 4) opción C $\rightarrow 59$

$$T(n) = \begin{cases} 1 & n = 1 \\ 2T(n/2) + 5n + 1 & n > 2 \end{cases}$$

$$T(4) = 2T\left(\frac{4}{2}\right) + 5 \cdot 4 + 1 =$$

$$= 2T(2) + 20 + 1 =$$

$$= 2 \cdot 19 + 20 + 1 =$$

$$= 38 + 20 + 1 = 59$$

$$T(2) = 2T\left(\frac{2}{2}\right) + 5 \cdot 2 + 1 =$$

$$= 2T(1) + 10 + 1 =$$

$$= 2 \cdot 1 + 10 + 1 = 19$$